Transmission Line Parameters

Class Notes

Notes by BALAJI SIR

DAY-2 [9/9/25] Transmission Line Performance

* By using a transmission line, electrical power is transferred from the remote generating station up to the load centre.

Aluminium Conductor Steel Reinforced (ACSR)

$$\text{No. of strands: } N = 3x^2 - 3x + 1$$

$$\text{Total diameter } D = (2x - 1)d$$

x = layer number , d = diameter of each strand.

$x = 3$

$$N = 3 \times 3^2 - 3 \times 3 + 1 = 19$$

$$D = (2 \times 3 - 1)d = 5d$$



ACSR conductor: 19 strands, D = 5d

The technical name of ACSR conductors are specified as animal names (Zebra, Panther, Lynx etc.) as per European standards and birds names (Turkey, Pigeon).

* Classification of Tr. line.

. Short line $l < 80$ km , $If < 4000$

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. Medium line $80 \leq l \leq 200$ km, $4000 \leq lf \leq 10000$

. Long line $l > 200$ km, $lf > 10000$

l = length of Tr. line, f = supply freq = 50 Hz

Eg: Short line $l \times f = 80 \times 50 = 4000$

. Communication line operating at MHz range, hence considered as long Tr. line because $lf > 10000$ for any length of line.

Eg: A T/L is excited by the frequencies of (1) 300 Hz (2) 50 Hz (3) 600 Hz. Find the type of T/L if its physical length is 20 km.

(1) $20 \times 300 = 6000$ - Medium

(2) $50 \times 20 = 1000$ - Short

(3) $20 \times 600 = 12{,}000$ - Long

Tr. line Parameters:

L - Due to flux linkage between the conductors.

C - Capacitance is due to potential difference between ACSR conductors.

Resistance : ACSR material itself has resistance.

Conductance : leakage current flowing through the supporting insulators is represented by conductance (G).

Inductance :

Skin and proximity effect are neglected.

Steps for calculation of Inductance

(1) $\oint H \cdot dx = I$

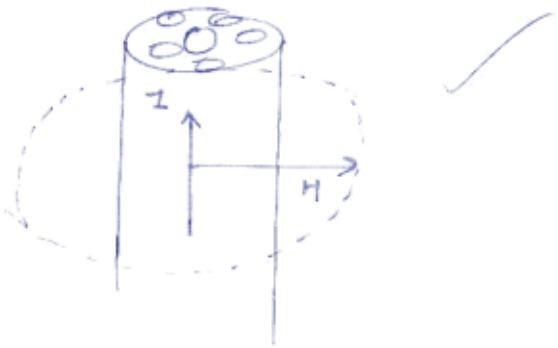
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(2) $B = \mu_0 \mu_r H$

(3) $\phi = B \cdot A$

(4) $\lambda = n\phi$

(5) $L = \frac{\lambda}{I}$



Current I in a conductor and magnetic field H around it

L of single ACSR conductors :

$$L_{ACSR} = L_{int} + L_{ext}$$

$$L_{int} = \frac{\mu_r}{2} \times 10^{-7} \text{ H/m}$$

$\mu_r \approx 1$ for ACSR

$$\boxed{L_{int} = \frac{1}{2} \times 10^{-7} \text{ H/m}}$$

$= 0.05$ mH/km

L_{int} is a constant and is independent of size of conductor because flux density and linkage is increasing from centre towards the surface linearly in proportion to current.

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External flux: conductor of radius r, distance d

$$L_{ext} = \frac{\mu_0 \mu_r}{2\pi} \ln\left(\frac{d}{r}\right) \; H/m$$

$\mu_r = 1$, $\mu_0 = 4\pi \times 10^{-7}$

$$\boxed{L_{ext} = 2 \times 10^{-7} \ln\left(\frac{d}{r}\right) \; H/m}$$

$$= 0.2 \ln\left(\frac{d}{r}\right) \; mH/km$$

$$\# \; L_{ACSR} = L_{int} + L_{ext} = 0.05 + 0.2 \ln\left(\frac{d}{r}\right) \; mH/km \; OR$$

$$= \frac{1}{2} \times 10^{-7} + 2 \times 10^{-7} \ln\left(\frac{d}{r}\right)$$

$$= 2 \times 10^{-7} \left[\frac{1}{4} + \ln\left(\frac{d}{r}\right)\right]$$

$$= 2 \times 10^{-7} \ln\left(\frac{d e^{1/4}}{r}\right)$$

$$= 2 \times 10^{-7} \ln\left[\frac{d}{r e^{-1/4}}\right]$$

$e^{-1/4} = 0.7788$, $r' = 0.7788 r$

$$\boxed{L_{ACSR} = 2 \times 10^{-7} \ln\left(\frac{d}{r'}\right)} \; H/m$$

Hollow conductor : $L_{int} = 0$

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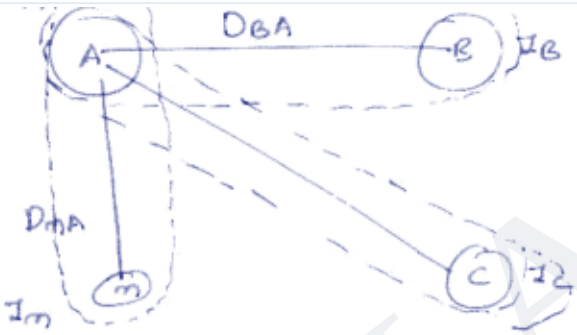
$$L_{Hollow} = 2 \times 10^{-7} \ln\left(\frac{d}{r}\right)$$

$$L_{Hollow} < L_{ACSR}$$

$$P_{Hollow} > P_{ACSR}$$

$$P \propto \frac{1}{L}$$

Flux linkage to a conductor in a group of conductors



Conductors A, B, C ... m carrying currents IA, IB, IC ... Im; spacings DBA, DCA, DmA

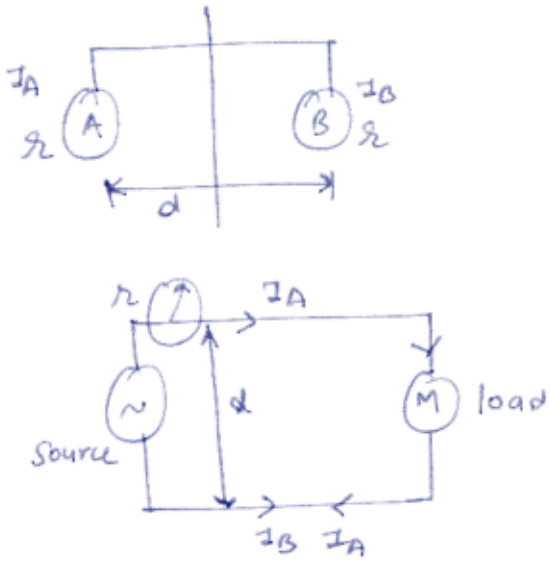
Flux linkage with A

$$\lambda = \frac{\mu_0}{2\pi}\left[I_A \ln\left(\frac{1}{r'}\right) + I_B \ln\frac{1}{D_{BA}} + I_C \ln\frac{1}{D_{CA}} + \cdots + I_m \ln\frac{1}{D_{mA}}\right]$$

At balance $I_A + I_B + \cdots + I_m = 0$

Case 1: L of 1-ϕ 2-wire T/L

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Two conductors A, B (radius r, spacing d) carrying IA and IB; source and load

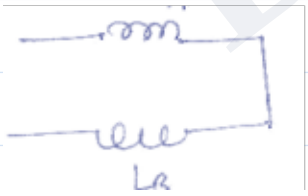
$$\lambda_A = \frac{\mu_0}{2\pi}\left[I_A \ln\left(\frac{1}{r'}\right) + I_B \ln\left(\frac{1}{d}\right)\right]$$

$$I_B = -I_A$$

$$\lambda_A = \frac{\mu_0 I_A}{2\pi} \ln\left(\frac{d}{r'}\right)$$

$$L_A = \frac{\lambda_A}{I_A} = \frac{\mu_0}{2\pi} \ln\left(\frac{d}{r'}\right) = \frac{\lambda_B}{I_B} = L_B$$

Total circuit / loop Inductance



Loop inductance: LA in series with LB

$$L_T = L_A + L_B = \frac{\mu_0}{\pi} \ln\left(\frac{d}{r'}\right) \text{ H/m}$$

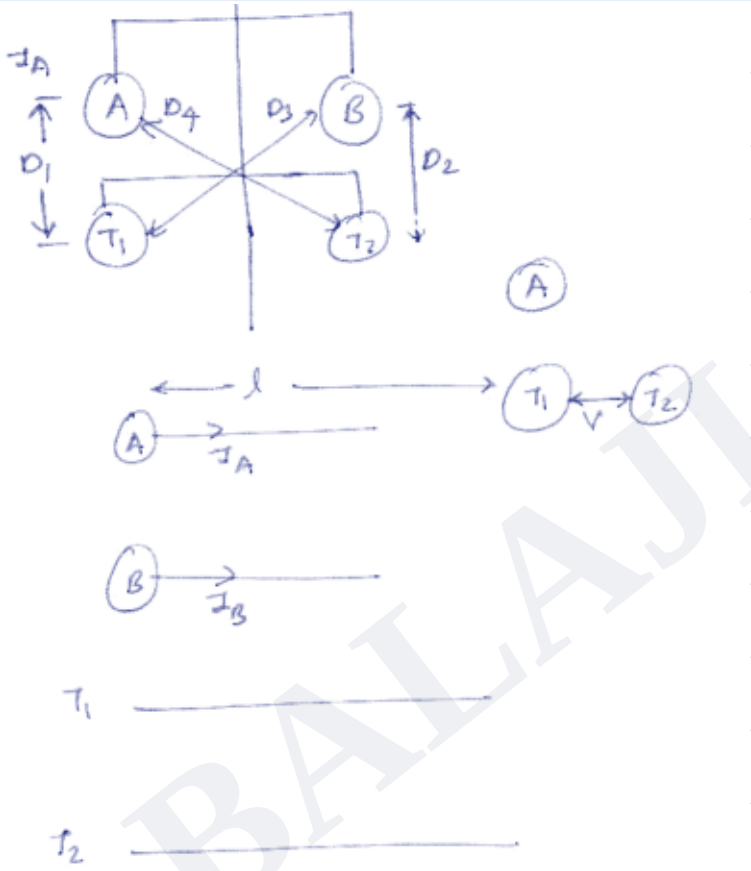
$$\boxed{L_T = 4 \times 10^{-7} \ln\left(\frac{d}{r'}\right) \text{ H/m}}$$

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Radio Interference in the communication line

If a power and telecommunication line are running close to each other, due to electrostatic and electromagnetic flux linkage additional vtg is induced in the telecommunication line conductors. It will produce disturbance to the Teleco...

(a) Untransposed line :



Power line A, B (IA, IB) and telecom line T1, T2; distances D1, D2, D3, D4; line length l; induced voltage V between T1 and T2

Flux linkage between T_1 and T_2

$$\lambda_{T_1} = \frac{\mu_0}{2\pi}\left[I_A \ln\frac{1}{D_1} + I_B \ln\frac{1}{D_3}\right]$$

$$\lambda_{T_2} = \frac{\mu_0}{2\pi}\left[I_A \ln\frac{1}{D_4} + I_B \ln\frac{1}{D_2}\right]$$

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$$\lambda_n = \lambda_{T_1} - \lambda_{T_2}$$

$$\lambda_n = \frac{\mu_0}{2\pi}\left[I_A \ln\frac{D_4}{D_1} + I_B \ln\frac{D_2}{D_3}\right]$$

At Balance $I_B = -I_A$

$$\lambda_n = \frac{\mu_0}{2\pi} I_A \ln\frac{D_4 D_3}{D_2 D_1}$$

Vtg induced between T_1 and T_2 due to current I_A and power frequency (f) is

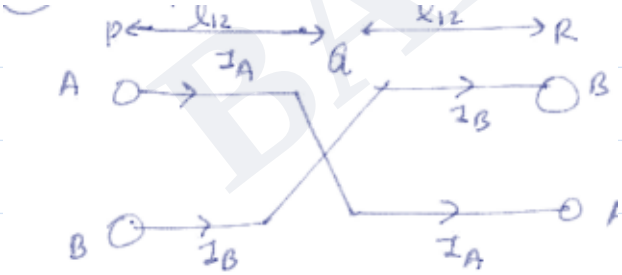
$$V = I_A X_L = I_A(2\pi f L_m) = 2\pi f \cdot \lambda_n$$

$$\boxed{V = \mu_0 f I_A \ln\left(\frac{D_4 D_3}{D_2 D_1}\right)}$$

$D_1 = D_2$ = shortest distance, $D_3 = D_4$ = longest distance (GATE)

$$\boxed{V = 2\mu_0 f I_A \ln\left(\frac{D_3}{D_1}\right)} \quad V/m$$

(b) Transposed lines



Transposed line: sections PQ and QR of length l/2 each

$$\lambda_{PQ} = \frac{\mu_0}{2\pi}\left[I_A \ln\frac{D_4}{D_1} + I_B \ln\frac{D_2}{D_3}\right]$$

$$\lambda_{QR} = \frac{\mu_0}{2\pi}\left[I_B \ln\frac{D_4}{D_1} + I_A \ln\frac{D_2}{D_3}\right]$$

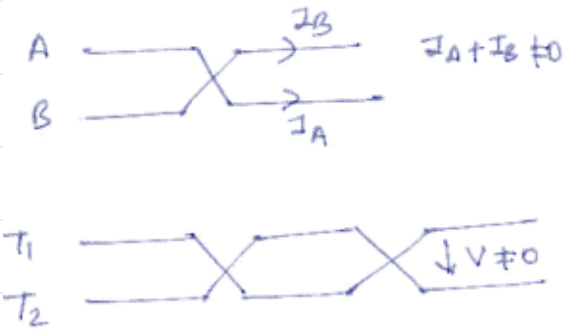
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$$\lambda_{avg} = \frac{\lambda_{PQ} \cdot \frac{l}{2} + \lambda_{QR} \cdot \frac{l}{2}}{l} = \frac{1}{2}(\lambda_{PQ} + \lambda_{QR})$$

$$\lambda_{avg} = \frac{1}{2}\frac{\mu_0}{2\pi}(I_A + I_B)\left[\ln\frac{D_4}{D_1} + \ln\frac{D_2}{D_3}\right]$$

At Balance $I_A + I_B = 0$

$\lambda_{avg} = 0 \Rightarrow V = 0$



Transposed power line (IA+IB is not zero) and transposed telecom line T1, T2 (V is not zero)

(c) Inductance per phase of a 3-ϕ Transposed power line.

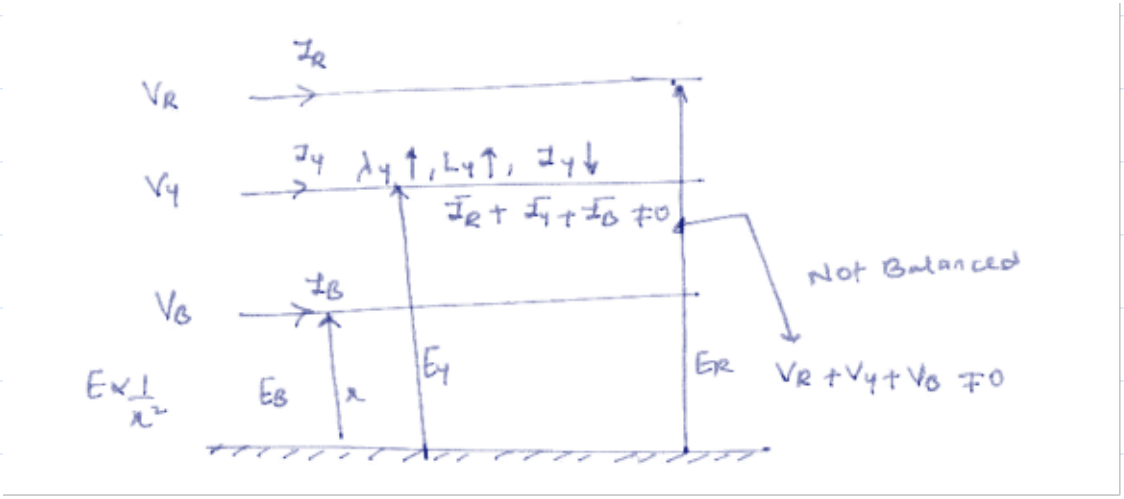
In case of untransposed power line, the middle conductor experiences maximum flux linkage causing an increase in inductance, so that current magnitude reduces and hence $I_R + I_Y + I_B \neq 0$ occurs. This is called electromagnetic Radio Interference.

* Conductor which is closest to Earth experiences maximum electrostatic induction. This will cause an increase in potential of the bottommost conductor so that $V_R + V_Y + V_B \neq 0$ occurs. This is called electrostatic interference.

* To eliminate or reduce this effect, a 3-ϕ transmission line is transposed at regular intervals of length of line. In India, practical transmission line is transposed for 220 kV and 400, 765 kV transmission line for every 100 km distance.

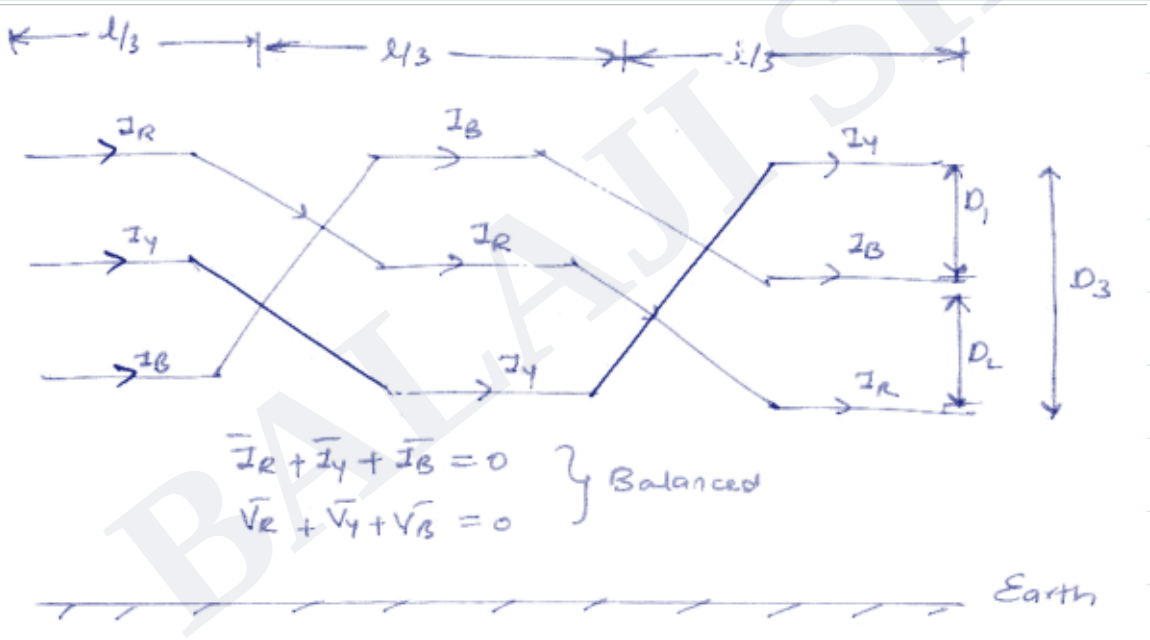
(a) Untransposed power line.

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Conductors VR, VY, VB above earth: ER, EY, EB (E proportional to 1/x^2); IR+IY+IB not balanced, VR+VY+VB not zero; lambda_Y up, L_Y up, I_Y down

(b) Transposed 3-ϕ T/line



Positions rotated every l/3 so each phase occupies every position; D1, D2, D3 spacings

$I_R + I_Y + I_B = 0$ } Balanced

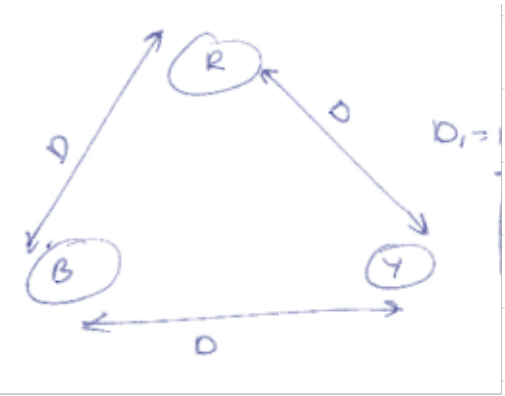
$V_R + V_Y + V_B = 0$

$$L_{ph} = L_R = L_Y = L_B$$

$$L_{ph} = 2 \times 10^{-7} \ln \left(\frac{D_1 D_2 D_3}{r'} \right)^{1/3}$$

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Equilateral Δ configuration



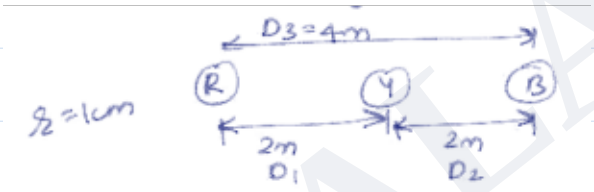
R, Y, B at the corners of an equilateral triangle, spacing D

$$D_1 = D_2 = D_3 = D$$

$$L_{ph} = 2 \times 10^{-7} \ln\left[\frac{D}{r'}\right]$$

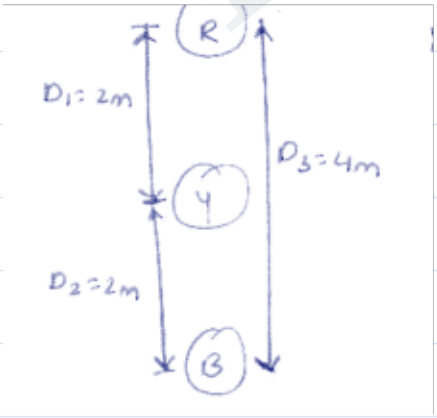
Eg: Find L_{ph}

(a) Horizontal configuration r = 1 cm



R, Y, B in a horizontal line: D1 = 2 m, D2 = 2 m, D3 = 4 m

(b) Vertical configuration



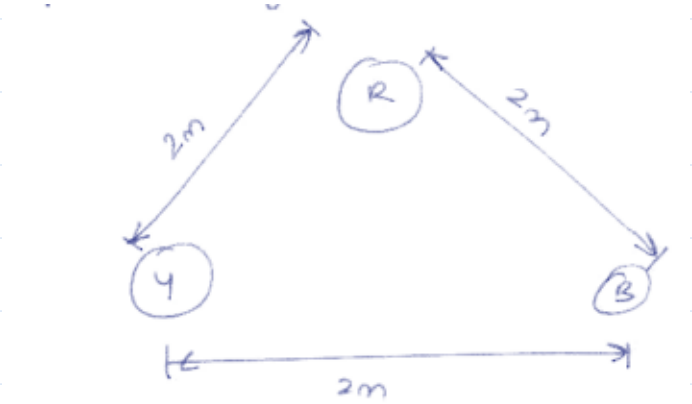
R, Y, B in a vertical line: D1 = 2 m, D2 = 2 m, D3 = 4 m

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$$L_{ph} = 2 \times 10^{-7} \ln \left(\frac{2 \times 2 \times 4}{0.7788 \times 1 \times 10^{-2}} \right)^{1/3}$$

$$= 1.15 \text{ mH/km/ph} \quad (L_H = L_V)$$

(c) Equilateral Triangular.



R, Y, B at corners of an equilateral triangle, 2 m sides

$$L_{ph} = 2 \times 10^{-7} \ln \left[\frac{2}{0.7788 \times 1 \times 10^{-2}} \right]$$

$$= 1.109 \text{ mH/km/ph}$$

$$= L_\Delta \quad [L_H = L_V]$$

$$P_\Delta > (P_H = P_V)$$

GMR and GMD :

1-ϕ 2-wire T/L



Conductors A, B of radius r at spacing d

$$L_A = L_B = 2 \times 10^{-7} \ln \left(\frac{d}{r'} \right) = 2 \times 10^{-7} \ln \left[\frac{D_m}{D_s} \right]$$

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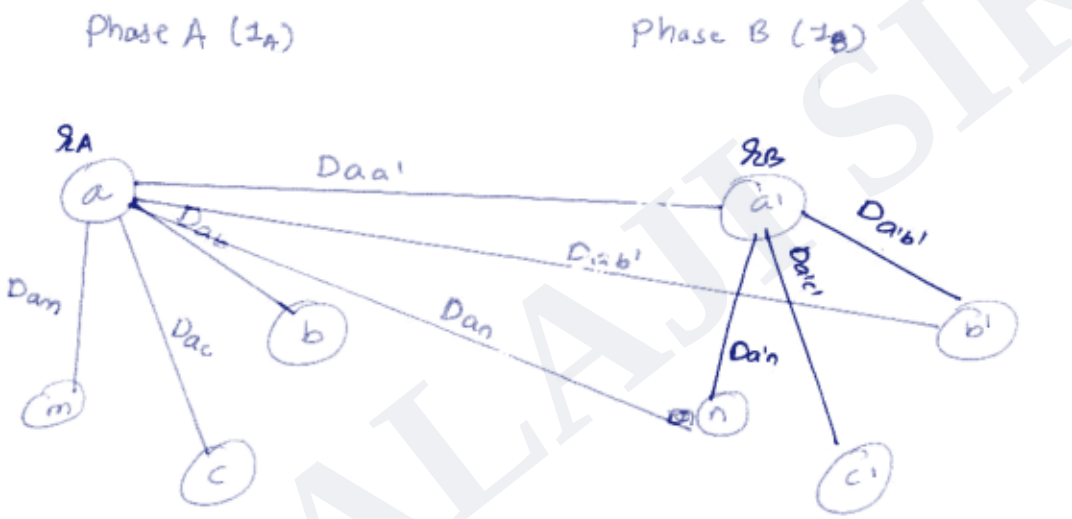
D_m = Mutual Geometric Mean Distance (GMD)

D_s = Self GMD = Geometric Mean Radius (GMR)

Inductance per phase of a bundled conductor

Bundled or composite conductor or group of ACSR conductors placed in each phase for sharing the load current, which will reduce the effective inductance and hence power transfer capability of the line improves.

2, 4, 6 bundled conductors configuration are practically used in 220, 400 and 765 kV HVAC systems and ±500, ±800 kV HVDC systems.



Phase A (IA): conductors a, b, c ... m; Phase B (IB): conductors a', b', c' ... n; mutual distances Daa', Dab', ...

$$L_A = 2 \times 10^{-7} \ln \left[\frac{D_{MAB}}{D_{SA}} \right] \text{ H/m/ph}$$

$$L_B = 2 \times 10^{-7} \ln \left[\frac{D_{MAB}}{D_{SB}} \right] \text{ H/m/ph}$$

$$D_{SA} = \left[(D_{aa} D_{ab} \cdots D_{am})(D_{ba} D_{bb} \cdots D_{bm}) \cdots (D_{ma} D_{mb} \cdots D_{mm}) \right]^{1/m^2}$$

where $D_{aa} = D_{bb} = \cdots = D_{mm} = r'_A = 0.7788 r_A$

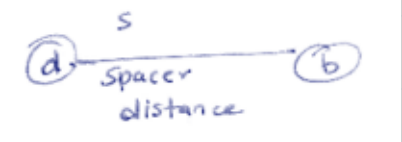
$$D_{SB} = \left[(D_{a'a'} D_{a'b'} \cdots D_{a'n})(D_{b'a'} D_{b'b'} \cdots D_{b'n}) \cdots (D_{na'} D_{nb'} \cdots D_{nn}) \right]^{1/n^2}$$

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where $D_{a'a'} = D_{b'b'} = \cdots = D_{nn} = r'_B = 0.7788 r_B$

$$D_{MAB} = \left[(D_{aa'} D_{ab'} \cdots D_{an})(D_{ba'} D_{bb'} \cdots D_{bn}) \cdots (D_{ma'} D_{mb'} \cdots D_{mn}) \right]^{1/(m \times n)}$$

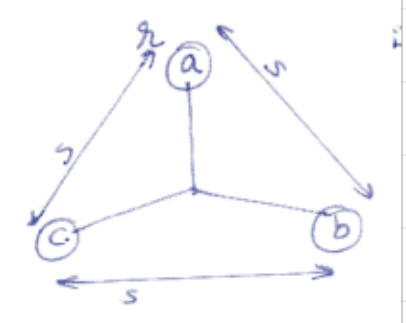
Eg: Bundle of (2)



Two sub-conductors a, b with spacer distance s

$$D_s = \left[(D_{aa} D_{ab} D_{ba} D_{bb}) \right]^{1/2^2} = (r' \cdot s \cdot s \cdot r')^{1/4} = (r' \cdot s)^{1/2}$$

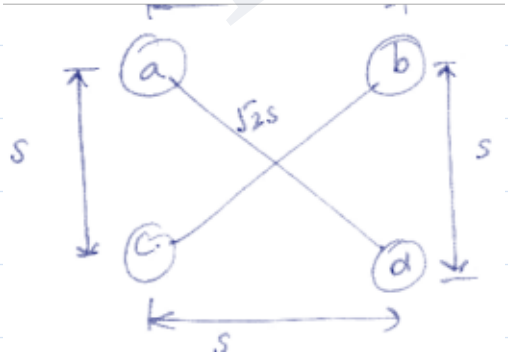
(b) Bundle of (3)



Three sub-conductors a, b, c at the corners of an equilateral triangle of side s

$$D_s = \left[(r' \cdot s \cdot s)(s \cdot r' \cdot s)(s \cdot s \cdot r') \right]^{1/3^2}$$

(c) Bundle of (4)



Four sub-conductors a, b, c, d at the corners of a square of side s; diagonal $\sqrt{2}\, s$

$$D_s = \left[(r' \cdot s \cdot \sqrt{2} s \cdot s)^4 \right]^{1/4^2} = (\sqrt{2} \cdot r' \cdot s^3)^{1/4}$$

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(d)

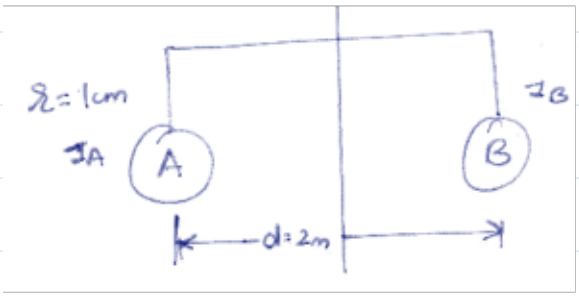


Three sub-conductors in a row: 1, 2, 3 (radius r)

$$D_s = [(r' \cdot 2r(4r)(2r \cdot r' \cdot 2r)(4r \cdot 2r \cdot r')]^{1/3^2}$$

Find the inductance per phase for the following configuration.

(1) with single conductor in each phase

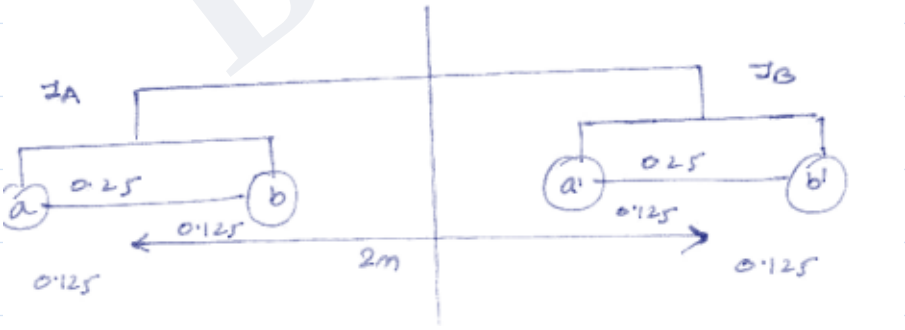


Single conductor per phase: r = 1 cm, d = 2 m

$$L_A = L_B = 2 \times 10^{-7} \ln\left(\frac{d}{r'}\right) = 2 \times 10^{-7} \ln\left(\frac{2}{0.7788 \times 1 \times 10^{-2}}\right)$$

$$= 1.109 \times 10^{-6} \text{ H/m/ph} = L_1$$

(2) with two bundle conductors in each phase.



Two sub-conductors per phase, spacing 0.25 m; phases 2 m apart

$$D_{SA} = D_{SB} = (r's)^{1/2} = (0.7788 \times 1 \times 10^{-2} \times 0.25)^{1/2} = 0.044 \text{ m}$$

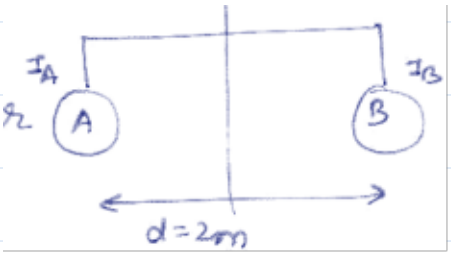
$$D_{MAB} = [(D_{aa'} D_{ab'})(D_{ba'} D_{bb'})]^{1/(2\times 2)} = (2 \times 2.25 \times 1.75 \times 2)^{1/4} = 1.99 \approx 2 \text{ m}$$

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$$L_A = L_B = 2\times10^{-7}\ln\left[\frac{D_{MAB}}{D_{SA}}\right] = 2\times10^{-7}\ln\left(\frac{2}{0.044}\right) \quad L_2 < L_1,\ P_2 > P_1$$

$$= 0.768\times10^{-6}\ H/m/ph = L_2$$

With single conductor of same power as that of bundle.

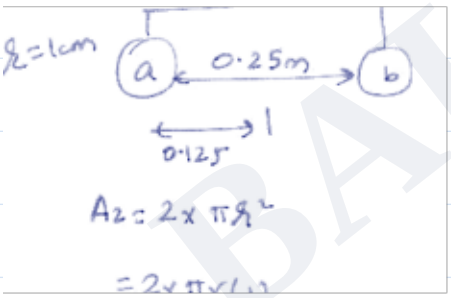


Single conductor per phase of radius r, d = 2 m

$$L_A = 2\times10^{-7}\ln\left[\frac{d}{r'}\right]$$

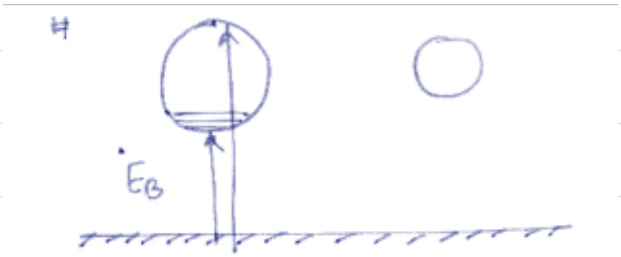
$$0.768\times10^{-6} = 2\times10^{-7}\ln\left[\frac{2}{0.7788\times r}\right] \Rightarrow r = 5.56\ cm$$

$$A_1 = \pi r^2 \approx 100\ cm^2 \approx 16A_2$$



Bundle of two: r = 1 cm, spacing 0.25 m

$$A_2 = 2\times\pi r^2 = 2\times\pi\times(1) = 6.28\ cm^2$$



Conductors above earth with electric field EB

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Advantages of Bundle conductor

(1) Self GMD is higher, so that L and X_L are reduced and hence power transfer capacity increases. [$P \propto \frac{1}{L}$]

(2) Corona power loss is reduced.

(3) Radio interference is reduced.

(4) Surge impedance is reduced so that surge impedance loading is increased.

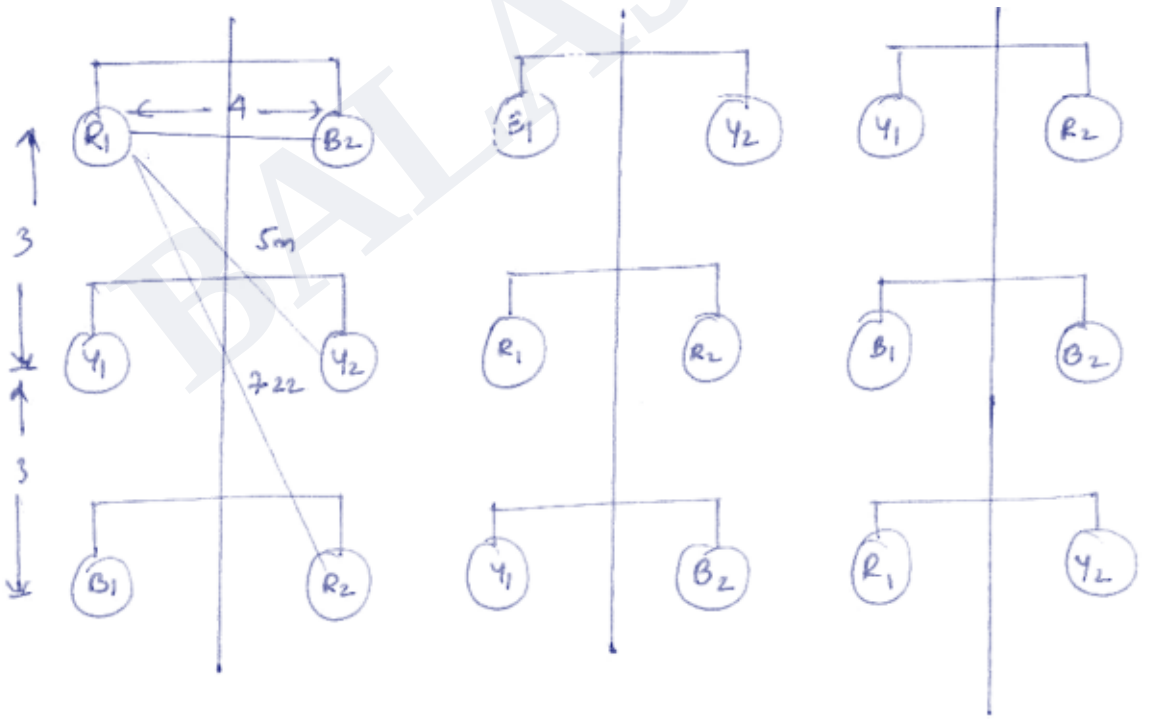
(5) For same power transfer capacity compared to single conductor, area and volume of conductor is reduced and hence cost of transmission is reduced.

(6) Electric field stress and vtg gradient is reduced compared to single conductor.

Because of above advantages bundled conductors are practically used in 220, 400, 765 kV and ±500, ±800 kV HVDC transmission.

DAY-3 [10/9/25] :

Case 5 : L_{ph} of a Double circuit line :



Double circuit tower: R1 B2 / Y1 Y2 / B1 R2 (with distances 4 m, 5 m, 7.22 m, 3 m); and the other two arrangements of the circuits (B1 Y2 / R1 R2 / Y1 B2, and Y1 R2 / B1 B2 / R1 Y2)

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In case of two bundled Tr. line $D_s = (r's)^{1/2}$, by increasing the spacer length, D_s can be increased so that inductance is reduced and hence power transfer capability is increased. These double circuits are practically used in 220, 400, 765 kV HVAC line and ±500, ±800 HVDC transmission.

-> To minimize the space occupancy of the electrical tower, instead of using two double circuits, one multi-circuit tower is used in which four circuits can be accommodated; circuits indicate R, Y, B together.

$$L_{ph} = 2 \times 10^{-7} \ln \left[\frac{D_{m1} D_{m2} D_{m3}}{D_{S1} D_{S2} D_{S3}} \right]^{1/3}$$

Cal of D_{S1} (R1, R2)

$$D_{S1} = [D_{R_1R_1} D_{R_1R_2} D_{R_2R_1} D_{R_2R_2}]$$

$D_{R_1R_1} = D_{R_2R_2} = r' = 0.7788 r$ = single

$= (r's)^{1/2}$ -> two bundle ; $= (r's^2)^{1/3}$ -> three ; $= (\sqrt{2}\, r's^3)^{1/4}$ -> four

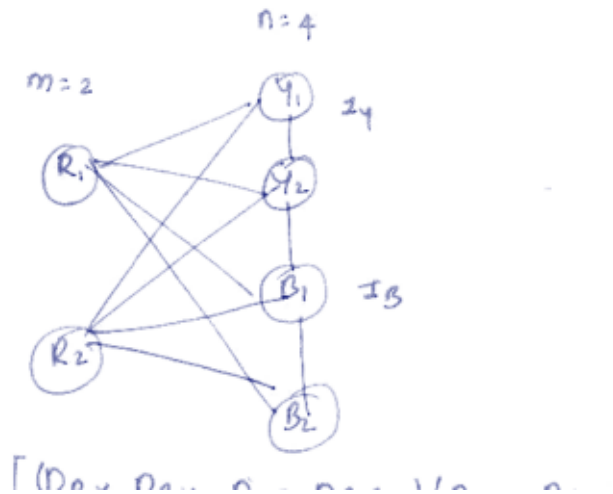
Similarly calculate D_{S2}, D_{S3}

$D_S = (r's)^{1/2} = (0.778 \times 1 \times 10^{-2} \times 7.22)^{1/2} = 0.237 \text{ m} = D_{S3}$

$D_{S2} = (0.7788 \times 10^{-2} \times 4)^{1/2} = 0.176 \text{ m}$

Calculation of D_{m1} $m = 2$, $n = 4$

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R1, R2 with Y1, Y2, B1, B2: m = 2 conductors of phase R, n = 4 conductors of the other two phases

$$D_{m1} = [(D_{R_1Y_1} D_{R_1Y_2} D_{R_1B_1} D_{R_1B_2})(D_{R_2Y_1} D_{R_2Y_2} D_{R_2B_1} D_{R_2B_2})]$$

Similarly calculate D_{m2}, D_{m3}

2 conductors :

$$D_{R_1R_1} = D_{R_2R_2} = (0.7788 \times 1 \times 10^{-2} \times 0.25)^{1/2}$$

$$D_{S1} = [0.044 \times 7.22 \times 7.22 \times 0.044]^{1/2^2} = 0.563$$

$$D_{m1} = [(3 \times 5 \times 6 \times 4)(5 \times 3 \times 4 \times 6)]^{1/2 \times 4} = 4.35 = D_{m3}$$

$$D_{m2} = [(3 \times 5 \times 3 \times 5)(5 \times 3 \times 5 \times 3)]^{1/2 \times 4} = 3.872 \text{ m}$$

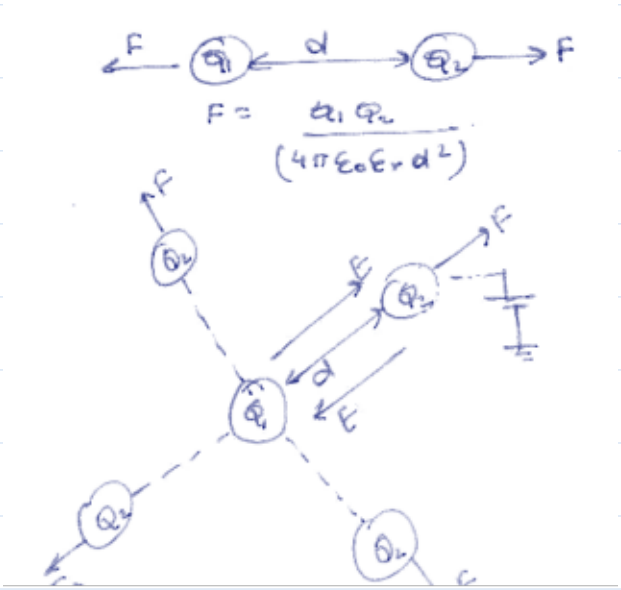
$$L_{ph} = 2 \times 10^{-7} \ln \left[\frac{(4.35)^2 \times 3.872}{(0.237)^2 \times 0.176} \right]^{1/3} = 5.94 \times 10^{-7}$$

$$= 0.594 \, \mu H/m = 0.594 \, mH/km/ph$$

Capacitance (C) :

Two charges Q_1, Q_2 at distance d : forces F on each; $F = \frac{Q_1 Q_2}{4\pi \varepsilon_0 \varepsilon_r d^2}$

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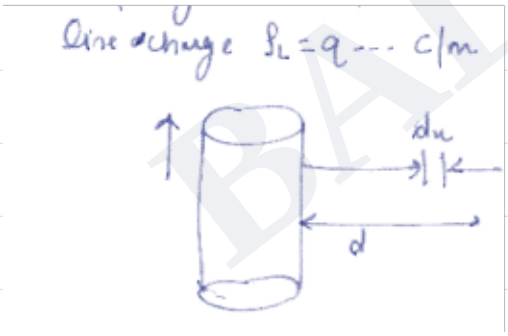
Charge Q1 surrounded by charges Q2, Q3, Q4, Q5; forces F

$$E = \frac{F}{Q_2} = \frac{Q_1}{4\pi\varepsilon_0\varepsilon_r d^2}$$

$$Vtg = dV = \frac{\text{work done}}{\text{Charge}} = \frac{\text{Force} \times \text{displacement}}{\text{Charge}}$$

$$dV = \frac{F}{Q}(dx), \quad dV = -E\,dx, \quad V = -\int_d^{} E \cdot dx, \quad C = \frac{Q}{V}$$

C of single ACSR conductor : line charge $\rho_L = q$ ... C/m



Cylindrical conductor, distance d, element dx

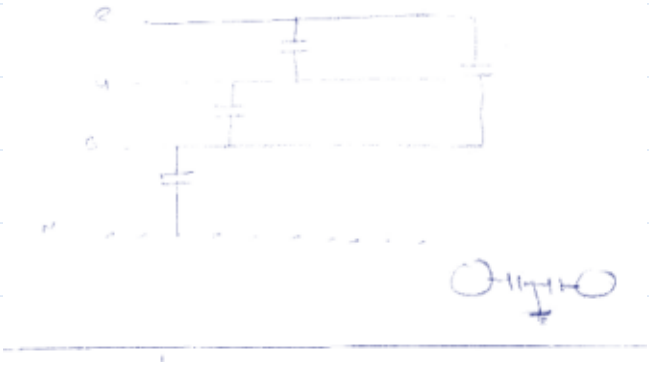
$$E = \frac{q}{2\pi\varepsilon_0\varepsilon_r x}$$

$$V = \frac{q}{2\pi\varepsilon_0\varepsilon_r} \ln\left(\frac{d}{r}\right)$$

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$$C_{ACSR} = \frac{q}{V} = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln\left(\frac{d}{r}\right)} \quad F/m$$

$$L_{ACSR} = \frac{\mu_0\mu_r}{2\pi}\ln\left(\frac{d}{r'}\right), \quad r' = 0.7788r$$



Capacitances between conductors and to earth

Configuration vs L_{ph} and C_{ph} :

1-ϕ wire :

$$L_{ph} = \frac{\mu_0\mu_r}{2\pi}\ln\left(\frac{d}{r'}\right), \quad C_{ph} = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln\left(\frac{d}{r}\right)}$$

Total / ckt / loop :

$$L = \frac{\mu_0\mu_r}{\pi}\ln\left(\frac{d}{r'}\right), \quad C_{AB} = \frac{\pi\varepsilon_0\varepsilon_r}{\ln\left(\frac{d}{r}\right)} = \frac{C_{ph}}{2}$$

Horizontal or Non-equilateral Triangular :

$$L_{ph} = \frac{\mu_0\mu_r}{2\pi}\ln\frac{(D_1D_2D_3)^{1/3}}{r'}, \quad C_{ph} = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln\frac{(D_1D_2D_3)^{1/3}}{r}}$$

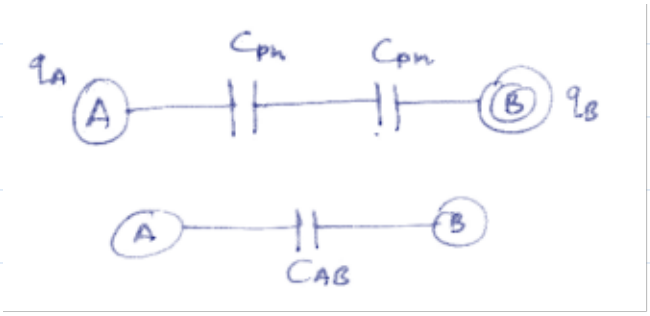
Bundle :

$$L_{ph} = \frac{\mu_0\mu_r}{2\pi}\ln\left(\frac{D_m}{D_s}\right), \quad C_{ph} = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln\left(\frac{D_m}{D_s}\right)}$$

Double ckt :

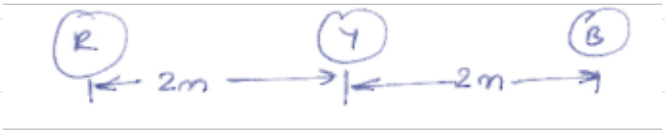
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$$C_{ph} = \frac{\mu_0 \mu_r}{2\pi} \ln\left(\frac{D_{m1}D_{m2}D_{m3}}{D_{s1}D_{s2}D_{s3}}\right)^{1/3}, \quad C_{ph} = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln\left(\frac{D_{m1}D_{m2}D_{m3}}{D_{s1}D_{s2}D_{s3}}\right)^{1/3}}$$



Capacitances Cph in series between A and B (charges qA, qB); equivalent CAB

(1) Horizontal

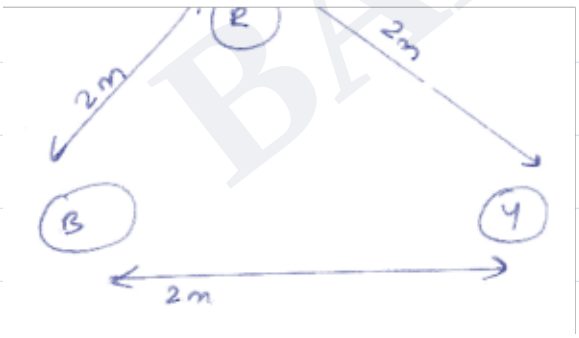


R, Y, B in a line, 2 m apart

$$C_{ph} = \frac{2\pi \times 8.854 \times 10^{-12}}{\ln\left[\frac{(2\times 2\times 4)^{1/3}}{1\times 10^{-2}}\right]} = 10.05\ pF/m/ph = C_H$$

$$L_H = 1.15\ mH/km/ph$$

(2) Equilateral Δ



R, Y, B at the corners of an equilateral triangle, 2 m sides

$$C_{ph} = \frac{2\pi \times 8.85 \times 10^{-12}}{\ln\left[\frac{2}{1\times 10^{-2}}\right]} = 10.49\ pF/m/ph = C_\Delta$$

$$L_\Delta = 1.109\ mH/km/ph$$

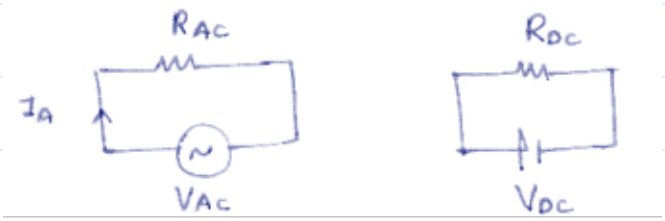
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$L_\Delta < L_H$

$P_\Delta > P_H$

$C_\Delta > C_H$

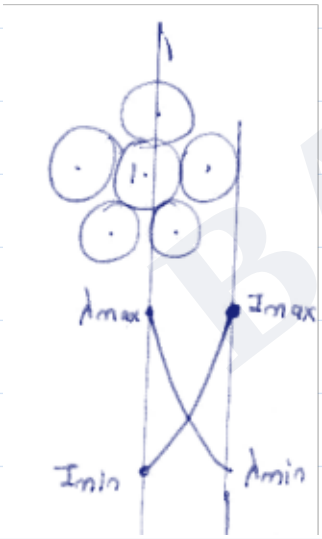
III Resistance (R)



AC circuit (RAC, VAC, IA) and DC circuit (RDC, VDC)

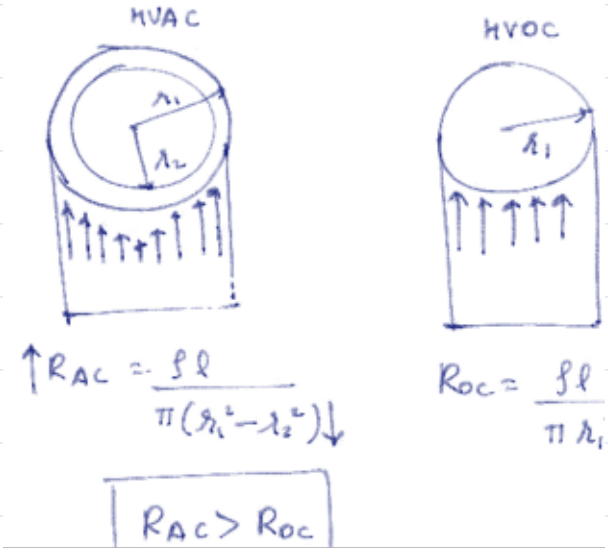
$R_{AC} > R_{DC}$ ($\approx 1.6 R_{DC}$) } (1) Skin effect (2) Proximity effect

(1) Skin effect



Current distribution across a stranded conductor: Imax and Imin, flux linkage lambda max and min

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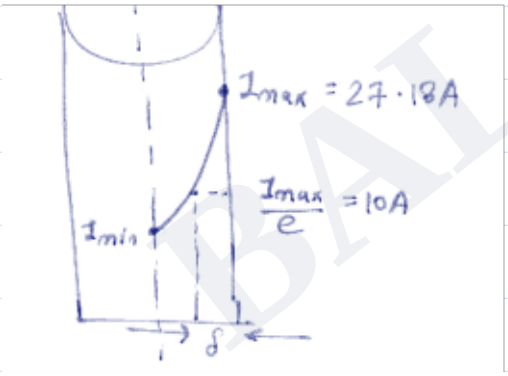


HVAC: current flows in the outer ring (r1, r2); HVDC: current through the full area r1

$$R_{AC} = \frac{\rho l}{\pi (r_1^2 - r_2^2)} \uparrow, \quad R_{DC} = \frac{\rho l}{\pi r_1^2}$$

$$\boxed{R_{AC} > R_{DC}}$$

Skin depth (δ) -> AC

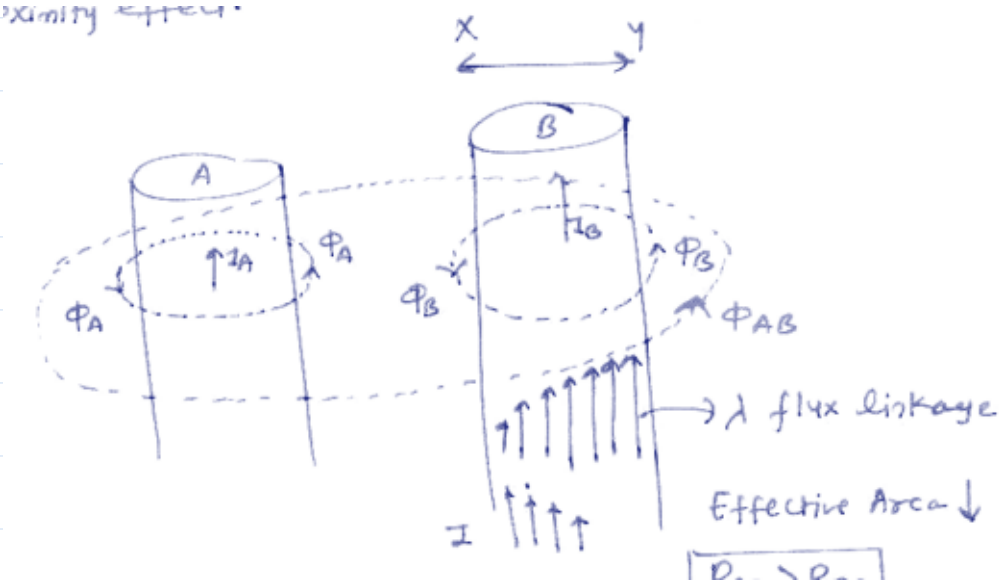


Imax = 27.18 A at the surface, Imax/e = 10 A at depth delta

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \mu_r \sigma}} \propto \frac{1}{(\text{skin effect})}$$

(2) Proximity effect.

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Conductors A and B carrying IA, IB with fluxes PhiA, PhiB, PhiAB; flux linkage lambda; effective area reduced

$$R_{AC} > R_{DC}$$

Skin effect :

* Accumulation of current on the surface of the conductor is called skin effect.
* Due to skin effect, effective current flowing path is reduced so that $R_{AC} > R_{DC}$ occurs.
* Skin effect depends on frequency, μ_r, σ, size of the conductor.
* Skin effect is more at higher frequency, so that this effect is more in communication line as compared to power line.
* At higher diameter conductor, skin effect is higher.

Proximity effect :

* Current flowing in the conductor produces mutual flux. This flux linkage becomes non-uniform in case of HVAC lines. This will cause non-uniform current flow and hence $R_{AC} > R_{DC}$ occurs.
* This effect depends on f, μ_r, σ and distance between the conductors.
* Power cables are very close to each other (R, Y, B) so that proximity effect is higher compared to overhead lines.

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(G) Conductance :

* Leakage current flowing through the supporting insulators is represented by conductance parameter (G).

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